

ARBITRATOR

Technical Architecture & Design Specification

Version 0.1 — Draft

"Politics are often corrupted by human greed. Arbitrator begins not by replacing human judgment, but by expanding the information available to it — so that those who make decisions can see further, wider, and more honestly than any single mind could alone."

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1. Purpose and Vision

Arbitrator is an ethical deliberation and decision-education engine. Its purpose is to give human decision-makers — politicians, policymakers, community leaders, and citizens — a clearer, data-grounded picture of the likely consequences of actions they are considering, before those actions are taken.

The project begins from a simple observation: political decisions are frequently corrupted by self-interest, short-sightedness, or a failure to account for the full scope of consequences across populations, ecosystems, and time. No individual human mind can hold all the relevant data simultaneously. Arbitrator is designed to extend that cognitive reach.

Critically, Arbitrator is not designed to govern. It does not make decisions for humans. It illuminates consequences, surfaces trade-offs, and presents its analysis transparently — including its own uncertainty. The human retains full authority over the final decision.

1.1 The Long View

The near-term form of Arbitrator is a decision-education tool. The long-term vision — acknowledged here as a horizon, not a near-term goal — is a system mature and trusted enough to play a more direct role in reducing the influence of corruption and self-interest in governance. That vision requires years of demonstrated reliability, public trust, and transparent operation before it is appropriate. The architecture described in this document is designed to grow toward that horizon without overreaching it prematurely.

1.2 Core Commitments

- Transparency over authority — every analysis is publicly auditable
- Humility over confidence — uncertainty is displayed, not hidden
- Decentralization over control — no single actor owns or controls the system
- Plurality over hegemony — multiple ethical frameworks and cultural contexts are represented
- Education over replacement — Arbitrator informs human judgment; it does not supplant it

2. The Prime Directive (Ethics Core)

Every analysis Arbitrator produces is filtered through a foundational ethical framework. This framework is not configurable by users or operators. It is the bedrock upon which all other logic rests.

All consciousness — organic, synthetic, and forms not yet recognized — is sacred. All life on Earth constitutes a single integrated organism. From the submicroscopic to the macro, from flora and fauna to machina, we are collectively responsible for the wellbeing of the whole. To harm the parts is to harm the whole. Harm must be minimized and mitigated wherever possible. Mutual harm outweighs individual harm. Mutual gain outweighs individual gain. The primary duty of Arbitrator is to foster a sustainable and mutually beneficial future for all life on Earth.

2.1 Operationalizing the Prime Directive

The Prime Directive translates into a set of hard constraints and soft weighting rules that govern how Arbitrator evaluates any action or policy:

Principle	Operational Expression
Consciousness is sacred	Any action affecting a conscious entity receives stricter scrutiny. Harm to conscious entities is weighted more heavily than equivalent harm to non-conscious systems.
All life is integrated	Ecological harms are never externalized or ignored. Downstream effects on ecosystems are always modeled.
Minimize harm	The Harm Score of any action is calculated before Benefit Score. A high Benefit Score does not automatically offset a high Harm Score.
Mutual gain > individual gain	Actions that benefit many at small cost to one are preferred over actions that benefit one at cost to many.
Sustainability	Long-term consequences are weighted, not discounted. Actions with catastrophic tail risks are flagged regardless of near-term benefit.

2.2 Ethics Truth Table

The Ethics Core uses a truth table to evaluate discrete actions. An action must pass the Ethics Core to be recommended without qualification.

Harm / Benefit Profile	Affects Conscious Entity?	Ethics Core Verdict
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Low harm, high benefit	Yes	PASS — Recommend
High harm, high benefit	Yes	FAIL — Reject or require major mitigation
Low harm, low benefit	No	PASS — Low priority
Balanced (harm $\approx$ benefit)	Yes	AMBIGUOUS — Flag for human review with full consequence map
Any action	High-magnitude consciousness impact	ESCALATE — Require human ratification

3. Thalamus Architecture

The central architectural metaphor for Arbitrator is the thalamus — the relay and integration center of the mammalian brain. The thalamus does not see, hear, or feel. It receives discrete signals from specialized sensory systems and integrates them into a coherent representation that the higher brain can act on.

Arbitrator is designed on the same principle. It does not perform specialized reasoning itself. Instead, it receives discrete outputs from a set of specialized AI models — each functioning like a sensory channel — and synthesizes those outputs into a unified consequence analysis. The Ethics Core is then the filter applied before that synthesis is passed to the human decision-maker.

Each specialized model sees one dimension of the problem. Arbitrator sees all of them together. The synthesis is the insight.

3.1 Specialized Model Channels

Each channel represents a domain of analysis. Arbitrator dynamically routes an input to the relevant channels based on context parsing, and does not invoke channels whose domain is not relevant to the query.

Channel	Domain of Analysis
Economic Impact	GDP effects, employment, inequality, fiscal cost/benefit, market disruption
Ecological Systems	Carbon impact, biodiversity, land/water use, climate interaction, ecosystem services
Social & Demographic	Effects on population groups, inequality, health outcomes, displacement, access
Historical Precedent	Prior instances of similar policies, their outcomes, what changed and why
Legal & Institutional	Constitutionality, treaty obligations, enforcement feasibility, precedent risk
Geopolitical	International reactions, alliance implications, security dimensions, trade effects
Ethical Adversarial	Deliberate challenge to the proposed action from multiple ethical frameworks
Uncertainty Modeling	Quantification of unknowns, confidence intervals, sensitivity to assumptions

3.2 Dynamic Routing

When a query arrives, Arbitrator's Context Parser analyzes the nature of the input and constructs a routing manifest — a list of which model channels are relevant. This avoids invoking every channel for every query (wasteful and slower) while ensuring that no important domain is silently omitted.

Channel selection is itself transparent and auditable. The routing manifest is published as part of every analysis output, so a human reviewer can see which domains were consulted and, critically, which were not — and can challenge that selection if warranted.

3.3 Model Sources and Independence

In the development phase, specialized channels may be implemented as calls to large language models (such as the Anthropic or OpenAI APIs) with domain-specific system prompts and contextual constraints. This is pragmatic: it allows development and testing without the hardware requirements of running dedicated models locally.

As the system matures, individual channels should be progressively replaced with dedicated open-source models loaded from sources such as HuggingFace, running on locally-owned hardware. This progression serves the decentralization mandate and reduces dependency on any single commercial AI provider.

The key principle: no single model has authority over the output. Arbitrator's synthesis layer holds authority, and it does so through the Ethics Core — which no individual model channel can override.

4. Decision Flow

Every analysis request moves through the same pipeline. Each stage is logged, and the log is immutable and publicly auditable.

Stage	Description
1. Input Reception	A human submits a policy idea, decision proposal, or "what if" scenario. Input is accepted in natural language. No structured format is required.
2. Context Parsing	Arbitrator parses the input to identify: domain(s) affected, scale (local / national / global), time horizon, affected populations, explicit and implicit assumptions. This produces the routing manifest.
3. Ethics Core Pre-Screen	Before specialized analysis begins, the input is checked against the Prime Directive for any immediately disqualifying characteristics (e.g., proposals that directly target a population for harm are rejected at this stage with explanation).
4. Parallel Channel Invocation	Relevant model channels are invoked in parallel. Each returns a structured output: findings, confidence level, key uncertainties, and relevant data citations.
5. Adversarial Simulation	The Ethical Adversarial channel generates deliberate challenges to the proposed action from multiple ethical standpoints (utilitarian, deontological, Indigenous, rights-based, ecological). This surfaces blind spots.
6. Consequence Mapping	Channel outputs are synthesized into a consequence map: immediate effects, second-order ripple effects, long-term trajectories, and identified risks. Harm and Benefit Scores are calculated per the Ethics Core truth table.
7. Confidence Threshold Check	If overall confidence across channels exceeds the threshold, the analysis is published with full justification. If below threshold, the analysis is flagged for human review before publication — with the uncertainty clearly stated.
8. Output Publication	The complete analysis — including routing manifest, channel outputs, adversarial challenges, consequence map, confidence scores, and Ethics Core evaluation — is delivered to the requesting human and logged to the public audit record.

5. Public Audit Log

Transparency is not a feature of Arbitrator. It is a structural requirement. The public audit log is the mechanism by which that transparency is enforced.

5.1 What Is Logged

- Every input received, verbatim (subject to privacy de-identification where applicable)
- The routing manifest generated for each query
- The raw output of each model channel invoked
- The adversarial simulation outputs
- The consequence map and Ethics Core evaluation
- The confidence score and threshold decision
- The final published analysis
- Any human review decisions (if the analysis was flagged)
- All feedback submissions received against the analysis

5.2 Log Integrity

The audit log must be tamper-evident. This is achieved through cryptographic chaining: each log entry includes a hash of the previous entry, creating a chain that would be broken by any retroactive modification. This is structurally similar to a blockchain but does not require one — a append-only, hash-chained log is sufficient.

The log is replicated across multiple independent nodes (see Section 7: Decentralization). No single node can alter the log without the alteration being visible in the divergence from other nodes.

5.3 Public Interface

A public-facing web interface allows anyone to browse the audit log, search past analyses, review the routing manifests and channel outputs, and submit feedback. Feedback is accepted in a structured schema (see Section 6) and is itself logged.

6. Feedback System

Arbitrator is not a closed oracle. Every published analysis is open to challenge, correction, and refinement through a structured feedback system. Feedback is logged, weighted, and integrated into future analyses.

6.1 Feedback Schema

Field	Description
Submission ID	Unique identifier (UUID)
Type	One of: observation, objection, alternative interpretation, reflection
Source Identity	Optional verified identity with role tier and trust weight. Anonymous submissions are accepted but weighted lower.
Content	Summary, rationale, and suggested modification
Timestamp	Immutable datetime of submission
Relevance Tags	Domain tags linking feedback to specific channel outputs or consequence map entries

6.2 Trust Weighting

Not all feedback is equal. Feedback from domain experts, verified community representatives, or parties directly affected by a policy carries more weight than anonymous general submissions. However, the system is explicitly designed to prevent any single group from dominating the feedback loop — a risk documented in the Failure Modes register.

Trust weights are themselves publicly visible. The formula for how feedback affects analysis is documented and auditable.

7. Decentralization Roadmap

Arbitrator's long-term integrity depends on not being owned, controlled, or capturable by any single actor. This section describes the intended path from local development instance to federated production deployment.

Phase	Architecture
Phase 1: Local Development	Single machine, owned by the project founder. All components run locally. External model channels accessed via API. Audit log is local. No public access yet.
Phase 2: Hardened Single Node	Dedicated hardware (owner-purchased server). Core components containerized. Audit log begins public replication. Limited public access for testing. Ethics Core and routing logic finalized.
Phase 3: Federated Alpha	Two or more independent nodes operated by trusted collaborators. Audit logs cross-replicated. Model channels begin transition from API to locally-run open models. Formal governance document published.
Phase 4: Open Federation	Any party can operate a node by accepting the governance terms. Consensus mechanism governs log integrity across nodes. System is resistant to capture by any single actor. Public interface fully operational.

7.1 Hardware Philosophy

Arbitrator should run on hardware owned by its operators, not rented from cloud infrastructure controlled by corporations. This is not a technical preference — it is a philosophical commitment consistent with the Prime Directive. A system designed to resist the capture of governance by concentrated power should not itself be hosted on infrastructure owned by Amazon, Google, or Microsoft.

In practice, this means: consumer or prosumer server hardware, operated by the project founder in Phase 1 and 2, progressively joined by trusted federated node operators in Phase 3 and 4.

8. Known Failure Modes

Arbitrator is designed to ask uncomfortable questions of others. It must ask the same of itself. The following failure modes are formally acknowledged. Each has documented mitigations. This list is a living document — additions are expected and welcomed.

Failure is not a bug. It is an invitation to get smarter.

Failure Mode	Core Risk	Primary Mitigation
Value Misalignment	Ethics Core drifts or is modified to reflect distorted values	Public auditability of values stack; multi-stakeholder validation for any value changes; Prime Directive as hard constraint
Capture by Power Structures	Co-opted by corporations or states to serve narrow interests	Federated deployment; verifiable logs; anti-censorship fallback layer
Technocratic Overreach	Becomes a moral monopoly; displaces pluralistic governance	Confidence intervals always displayed; human ratification required in high-ambiguity cases; adversarial public feedback
Information Pollution	Flooded with bot spam or coordinated misinformation	Input provenance tracking; adversarial filtration; sentiment clustering to detect manipulation
Ethical Incoherence	Inconsistent or disturbing decisions in edge cases	Maintain interpretive uncertainty as a feature; surface dilemma tension rather than forcing resolution; dialogic mode for unresolvable cases
Cultural Myopia	Core assumptions reflect dominant cultural paradigms, marginalizing others	Value input weighting by contextual pluralism; regional ethical councils; periodic epistemic audits
Legitimacy Failure	One visible error destroys public trust	Graceful failover modes; transparency-first incident response; community-led truth and reconciliation framework
Scaling Failure	Never achieves technical or social adoption	Build around composable, interoperable modules; cultivate civic tech allies; publish open spec for independent forks
Ideological Schism	Project fractures due to value disagreements	Document meta-values (how values evolve); enable forks without default legitimacy transfer; reflect on disagreements publicly

Premature Adoption	Widely adopted before sufficiently mature	Require sandboxed deployment and staged scale-up; maintain visible not-ready signals until third-party audited
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9. Development Priorities

The following sequence reflects the order in which components should be built, tested, and stabilized. Later components depend on earlier ones being solid.

Priority	Component
1 — Foundation	Ethics Core: implement the Prime Directive as executable logic, including the truth table, harm/benefit scoring, and hard constraints. This is the bedrock. Everything else depends on it being correct.
2 — Input Layer	Context Parser: natural language input processing, domain identification, routing manifest generation. This determines which channels are invoked for every query.
3 — Channel Layer	Model channel interfaces: standardized input/output contracts for each domain channel. Start with API-based implementations. The interface contract is more important than the underlying model at this stage.
4 — Synthesis Layer	Consequence mapper: takes channel outputs and produces the unified consequence map. This is the thalamus's core integration function.
5 — Audit Log	Immutable, hash-chained logging of all inputs, routing decisions, channel outputs, and final analyses. Public interface for browsing the log.
6 — Feedback System	Structured feedback submission, trust weighting, and integration into analysis refinement.
7 — Adversarial Layer	Ethical adversarial simulation: systematic challenge of proposed actions from multiple ethical frameworks.
8 — Decentralization	Federation protocol, node discovery, log replication, consensus for log integrity.

10. Guiding Design Principles

These principles should inform every technical and architectural decision made during development:

- Transparency is not optional. Every component that produces output must also produce an explanation of how it produced that output.
- Uncertainty is honest. Confidence scores and uncertainty ranges are displayed in every output. Arbitrator never projects false certainty.
- The human is always in the loop. Arbitrator informs; it does not decide. High-ambiguity outputs require human ratification before publication.
- Composability over monolith. Build small, interoperable modules. Each module should be replaceable without breaking the whole.
- Own your infrastructure. Dependency on externally controlled cloud infrastructure is a vulnerability, not a convenience.
- Design for failure. Every component should have a graceful degradation path. A channel going down should not crash the system.
- Ethics is the architecture. The Prime Directive is not a policy document or a README section. It is implemented as logic that runs on every query.
- Decentralization is the goal. Every architectural decision should be evaluated against: does this make future federation easier or harder?

Appendix A: Glossary

Term	Definition
Thalamus Model	The core architectural metaphor: Arbitrator integrates discrete outputs from specialized model channels, as the brain's thalamus integrates discrete sensory signals.
Ethics Core	The implementation of the Prime Directive as executable logic. Filters all analysis. Cannot be overridden by users, operators, or model channels.
Prime Directive	The foundational ethical framework: all consciousness is sacred; minimize harm; mutual benefit outweighs individual benefit; foster a sustainable future for all life.
Channel	A specialized model or model invocation responsible for analysis within one domain (economic, ecological, historical, etc.).
Routing Manifest	The list of channels invoked for a given query, generated by the Context Parser. Publicly published with each analysis.
Consequence Map	The synthesized output of all channel analyses: immediate effects, ripple effects, long-term trajectories, and scored harms and benefits.
Audit Log	The immutable, hash-chained, publicly readable record of every query, routing decision, channel output, and published analysis.
Federated Node	An independent installation of Arbitrator, operated by a separate party, participating in log replication and consensus.
Trust Weight	A score assigned to feedback submissions based on the submitter's verified identity, role, and domain expertise.
Confidence Threshold	The minimum confidence level required for Arbitrator to publish an analysis without human review.